EAST WEST UNIVERSITY

Project Report

Title: Creating a 3D Bulb with Filament in AutoCAD

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Course Code: CSE 200

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Introduction

This report described the creation of a 3D bulb with filament using AutoCAD. It includes detailed steps and dimensions to guide others in replicating the model.

Description of Geometric Units

The dimensions of the 3D bulb and its components are as follows:

1. Bulb Glass:

- Height: 127 m
- Maximum Diameter: 50 m

2. Screw Base:

- Height: 32 m

3. Filament:

- Length: 96 m

These dimensions are approximate and can be adjusted as needed.

Step-by-Step Drawing Process

❖ Creation of Bulb Glass

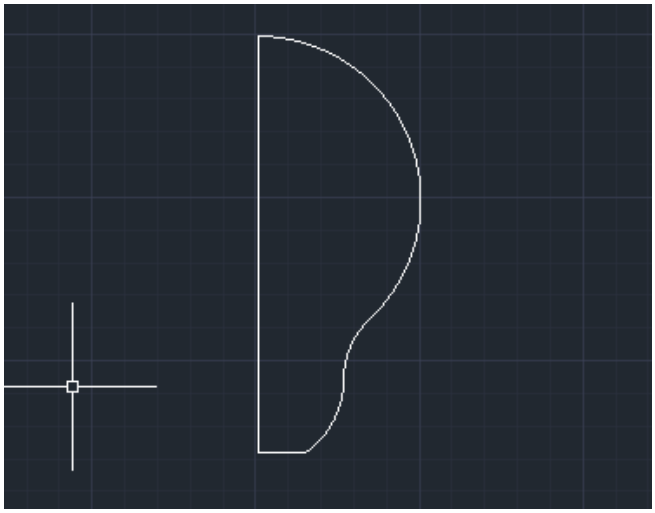
1. Draw the Profile:

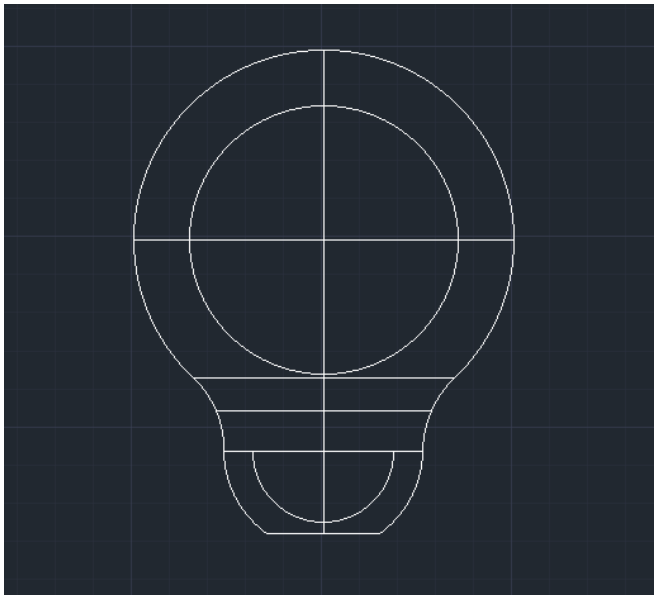
Use the Line and Circle commands to create a 2D profile of the bulb.

Ensure the height is 127 m and the maximum diameter is 50 m.

2. Revolve the Profile:

Select the profile and use the revolve command around the vertical axis to create the 3D bulb shape.





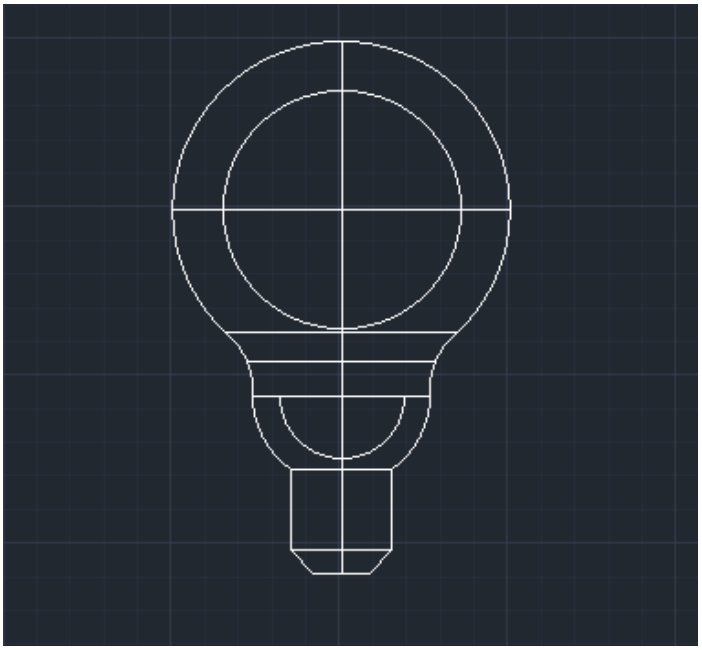
- **Designing the Screw Base**

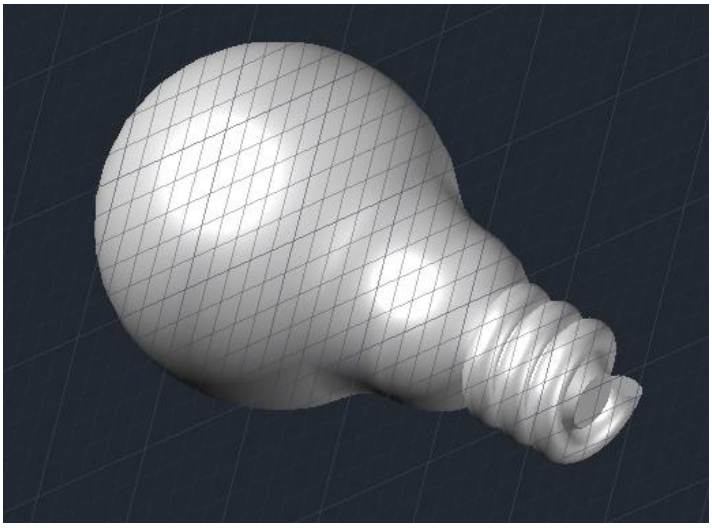
1. Draw the Base:

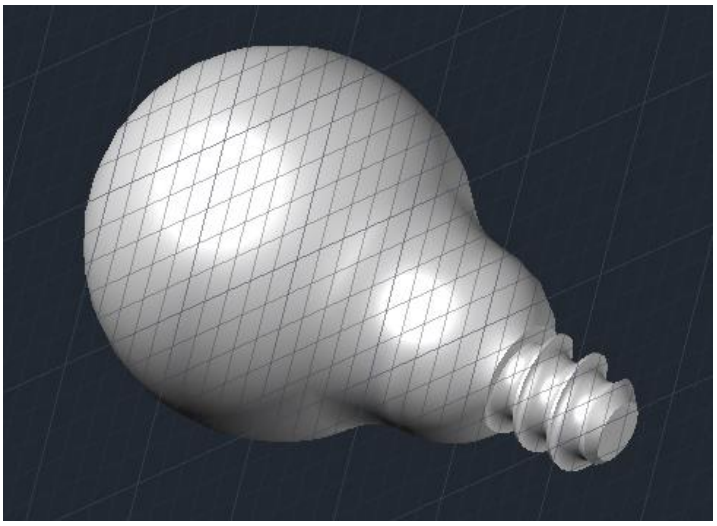
Use the polyline command to create a base and use the revolve command around the vertical axis to create the 3D. Then use the shaded command.

2. Add Threads:

Use the helix command for the thread path and the sweep, subtract command to create the threads along the helix.







- **Modeling the Filament**

1. Create a Shape:

Use the plain command to create a shape of filament. Then use the revolve command around the vertical axis to create the 3D.

2. Convert Spiral into Solid:

Use the sweep command to sweep a circular profile along the spiral path.

- **Final Assembly**

1. Position the Components:

Use the move commands to place the filament correctly within the bulb and apply the color red to the body of the filament.

2. Combine Components:

Use the union commands to assemble the parts seamlessly.

Conclusion

This project highlights the process of creating a 3D bulb with filament in AutoCAD, demonstrating the modeling techniques required. Detailed steps ensure that the model is replicable by others.

5. References

- AutoCAD User Manual
- Online Tutorials on 3D Modeling

THANK YOU